

Mobile Developer This term refers to individuals specializing in developing mobile (smartphone) applications using programming languages such as...Java, Kotlin, Objective-C, Swift, JavaScript Popular applications today are developed and released through app stores such as Google Play for Android and the App Store for iOS.



With the rapid growth of the smartphone market today, the mobile programming industry has become very popular, offering many job opportunities with good compensation. So, what are the reasons? **What do mobile developers need to learn?** In order to meet the needs of employers, and to be able to create good applications that attract many downloads and users.

The first Mobile app development involves developing applications for mobile devices, so you will get... **frequently working with smartphone devices** From iPhones to iPads, from the Z Fold to the Z Flip, or more simply, the pure Google Pixel line; and of course, sometimes these are virtual devices (Simulators or Emulators).

Monday The interface of the applications you write will be confined to a screen space of about 5-7 inches, which is much more limited than creating desktop or web

applications; this makes you **Certain design skills and UI/UX understanding are required to optimize the user interface and experience.**

Tuesday, Working with sensors, peripherals, and connections. Despite their compact size, smartphones are fully equipped with features such as cameras, speakers, fingerprint sensors, and facial recognition. Furthermore, modern smartphones can do almost everything: from payments and shopping to location tracking and printer connectivity – so many things that programmers have to learn to work with them.

Final Mobile applications need to be released through marketplace platforms like Google Play or the App Store to reach users, so you will also get... **Regularly familiarize yourself with the process of deploying applications to those stores.** New operating systems are updated frequently; once a year, the operating system is upgraded to a new version with relatively large changes to features and design. Therefore, your applications also need to be updated regularly.

First, to become a mobile programmer, you need to have knowledge of the programming field and skills for working in a team and independently, such as:

- Communication skills
- Foreign language skills
- Problem-solving skills
- Sense of responsibility towards work

To get started with mobile programming, begin by identifying the programming language and application platform you want to pursue from the outset. There are currently quite a few options available for writing mobile applications: ***mobile native, web-based app hay cross-platform.***

- **Mobile native:** Using programming languages created by the founders of the operating systems: Google Android provides developers with Android Studio, which can be written in Java or Kotlin; Apple iOS provides the Xcode IDE, which can be written in Objective-C or Swift. If you are just starting out in programming, choosing one language and going natively will help you go far in this field.
- **Cross-Platform and Hybrid (Web-based is a type of Hybrid):** Using other programming languages like JS, HTML5, Dart, and C# with frameworks like React, Flutter, and Xamarin helps develop applications that can run simultaneously on multiple platforms. This approach is more suitable for those who want to try mobile programming but already have a foundation in

other languages. For example, someone with experience in ReactJS can completely use React Native to develop their mobile application.

Acquire knowledge of UI/UX design.

Mobile applications always have limitations in display space, but there is a great diversity of screen types and devices, therefore requiring consistency in UI layout and user-friendly UX design.

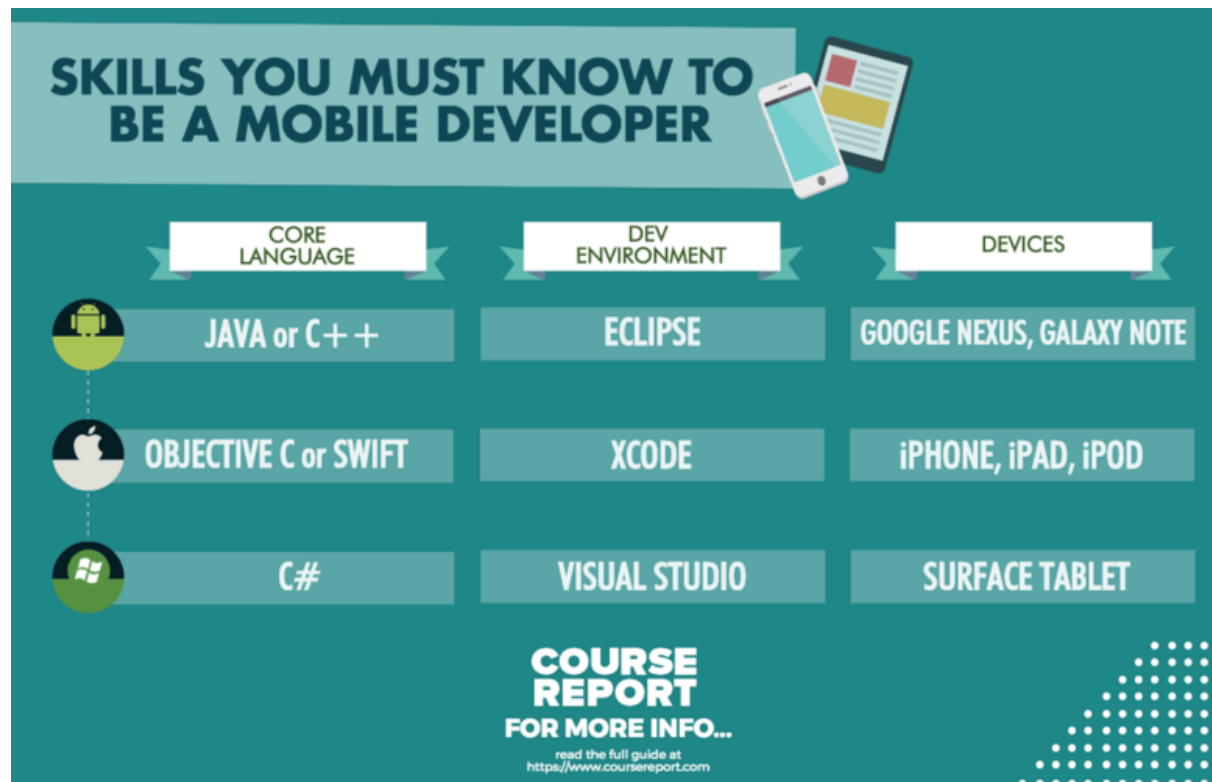
Key design elements to understand for mobile app design:

- Navigation, the movement of blocks, between screens
- Touchscreen, buttons, display area
- Font, color, consistency across screens
- Data entry forms collect information.

Write your own automated tests.

Any application created needs to be tested before release. The same applies to mobile applications; having automated testing capabilities is crucial if you want your application to be polished and complete by the time it reaches users. ***Here are a few tools you can consider using to write automated testing software::***

- Selendroid
- Appenium
- Espresso



Learn how to publish your app to the store.

As mentioned at the beginning of this article, publishing your application on app stores (App Store or Google Play) is essential for your application to reach a wider audience.

You may need to familiarize yourself with the following concepts beforehand:

- Với Google Play: Google Play Console, keystore, APK, App Bundle, ...
- Với App Store: Certificate, Provisioning, AppConnect, Apple Developer, ...